

Exam (2H00)

Exercise 1 Verify that the local balance equation,

$$\pi_i p_{ij} = \pi_j p_{ji}, i, j \in S,$$

holds for the Metropolis-Hastings algorithm.

Exercise 2 Consider the integral $I = \int_a^b H(x)dx$.

1. Propose an estimator $\widehat{I}_N$ of I using Monte Carlo method.
2. Let the estimator $\widehat{I}_{2N} = \frac{(b-a)}{2N} \sum (H(X_i) + H(b+a-X_i))$. Prove that if $H(x)$ is monotonic in x , then

$$\text{Var}(\widehat{I}_{2N}) \leq \frac{1}{2} \text{Var}(\widehat{I}_N).$$

Exercise 3 Apply the inverse-transform method to generate random variables of a Laplace distribution with probability density function

$$f(x) = \frac{\lambda}{2} \exp(-\lambda|x-\theta|), x \in \mathbb{R}, (\forall \theta \in \mathbb{R} \text{ and } \lambda > 0).$$

Exercise 4 A garage is equipped with a car wash. Experience shows that cars arrive according to a Poisson process with rate of 5 cars per hour. The time needed to wash a car is an exponential variable averaging 10 minutes per car. Only one car can be washed at a time. In addition, the garage has sufficient space to park all cars in queue.

1. Determine the model studied. Can the system reach steady state?
 - a) Find the mean number of cars in queue.
 - b) Calculate the mean waiting time of a car in the queue.
 - c) Give the proportion of time where the wash station is inactive.
 - d) Give the mean time spent in the garage.
2. Let's assume that the number of parking spaces, including the wash area inside the garage is limited to N .
 - a) Give the stationary distribution and the mean number of cars in the system as a function of N .
 - b) What is the number of positions N required to enable a driver to park inside the garage 8 times out of 10?
3. Let's assume that there are 8 parking positions in the garage (not including the wash area). When the garage is full, customers have to go to another carwash.
 - a) Calculate the number of customers lost (customers who can't get into the garage) ; assuming the number of hours worked is 8 per day.
 - b) Give the mean arrival rate $\bar{\lambda}$. Deduce the mean time spent in the garage.

Exercise 5 A workshop includes two automatic machines ($M1$ and $M2$) which are maintained by an operator. For machine $M1$, the running time is an $\exp(\lambda_1)$ variable; while for machine $M2$ this time is $\exp(\lambda_2)$. The operator can only repair one machine a time, and the repair time (for $M1$ and $M2$) follows $\exp(\mu)$. The two machines are of different importance in that a failure of $M1$ is more serious than a failure of $M2$. If $M1$ breaks down while $M2$ is being repaired, $M2$ is stopped and $M1$ is repaired immediately. As soon as $M1$ is up and running again, we resume the repair of $M2$. Let $\{X(t), t \geq 0\}$ a continuous time Markov chain with states space $\{0, 1, 2, 3\}$, where

- 0 : both machines are in service;
- 1 : $M1$ is being repaired, $M2$ is in service;
- 2 : $M2$ is being repaired, $M1$ is in service;
- 3 : $M1$ is being repaired, $M2$ is out of service.

(French translation: Un atelier comprend deux machines ($M1$ et $M2$) automatiques qui sont entretenues par un opérateur. Pour la machine $M1$, la durée de fonctionnement est une variable $\exp(\lambda_1)$; tandis que pour la machine $M2$ cette durée est $\exp(\lambda_2)$. L'opérateur ne peut réparer qu'une machine à la fois, le temps de réparation (pour $M1$ et $M2$) suit une loi $\exp(\mu)$. Les deux machines sont d'importances différentes en ce sens qu'une panne de $M1$ est plus grave qu'une panne de $M2$. Si $M1$ tombe en panne lors de la réparation de $M2$, on arrête cette dernière pour réparer tout de suite $M1$. Dès que $M1$ fonctionne à nouveau, on reprend la réparation de $M2$. Soit $\{X(t), t \geq 0\}$ une chaîne de Markov à temps continu et à espace d'états $\{0, 1, 2, 3\}$, où

- 0 : les deux machines sont en service ;
- 1 : $M1$ est en réparation, $M2$ est en service;
- 2 : $M2$ est en réparation, $M1$ est en service;
- 3 : $M1$ est en réparation, $M2$ est hors service.)

1. Give the representative transition graphe and transition matrix of this Markov chain.
2. Present the simulation algorithm for this continuous-time Markov chain.